BAPTISTE PLAQUEVENT-JOURDAIN

PERSONAL INFORMATION

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RESEARCH EXPERIENCE

Postdoc at CentraleSupélec (OPIS-CVN), Gif-sur-Yvette, France *03/2026-02/2027*
Nonsmooth differentiation by conservative mappings
for bilevel optimization with Antonio Silveti-Falls

PhD thesis in international co-supervision *01/2022-08/2025*
Université de Sherbrooke – Inria Paris
"A Robust Linearization Method for Complementarity Problems
A Detour Through Hyperplane Arrangements": manuscript (in French), slides
List of honor (2026) of the Faculty of Sciences of Université de Sherbrooke
With Jean-Pierre Dussault and Jean Charles Gilbert
Nonsmooth analysis/optimization, complementarity, computational geometry

Stage-CDD, Inria Paris, Paris, France *04/2021 – 12/2021*
Master internship and "Pre-thesis" on the PhD's topic

Gap year, Inria Saclay – École Polytechnique, Palaiseau, France *03/2020 – 08/2020*
Internship at RANDOPT: BFGS for multiobjective optimization,
Maximization of the hypervolume of the Pareto front

Gap year, Thales DMS, Élancourt, France *09/2019 – 02/2020*
Internship on modelling of meta-networks for wave redirection

Research Internship, Delft University, The Netherlands *05/2019 – 08/2019*
Finite Elements and Newton's method for nonlinear PDEs

STUDIES

Master 2 Optimization, Palaiseau-Orsay – IP Paris *2020 – 2021*
English formation in optimization, analysis, convexity,
optimal control, markov processes, game theory
Classes (September-April) then research internship (Ranked 1st/17)

ENSTA Paris, Palaiseau – general engineer degree *2017 – 2021*
Sciences of Optimization and Data specialization:
Linear optimization, graphs, metaheuristics, complexity
Applied Mathematics option (2nd year)

CPGE (MP), High School Chateaubriand Rennes *2014 – 2017*

PUBLICATIONS

- (published) *On the B-differential of the componentwise minimum of two affine vectorial functions*, Jean-Pierre Dussault, Jean Charles Gilbert, et B. P.-J., Math. Prog. Comp., 2025, Volume 17, pages 1–52 (detailed version).
- (accepted at GECCO'21) *DMS and MultiGLODS: black-box optimization benchmarking of two direct search methods on the bbob-biobj test suite*, Dimo Brockhoff, B. P.-J., Anne Auger, Nikolaus Hansen, 2021, pp. 1251–1258.

- (submitted to JMLR) *Differentiating Minimal-Norm Solutions to Parametric Optimization Problems*, B. P.-J., Jalal Fadili and Antonio Silveti-Falls (preprint).
- (in preparation) *Primal and dual approaches for the chamber enumeration of real hyperplane arrangements*, Jean-Pierre Dussault, Jean Charles Gilbert, and B. P.-J. (detailed version, two subsequent papers are in preparation).

SOFTWARE

- `ISF.m` : code and documentation of *On the B-differential of the componentwise minimum of two affine vectorial functions*, identification of the chambers of central arrangements and related problems.
- `ISF.jl` : code and documentation of *Primal and dual approaches for the chamber enumeration of real hyperplane arrangements*, identification of the chambers of arrangements.

EDUCATIONAL ACTIVITIES

Optimization Teaching at ENSTA Paris and Sherbrooke *2021-2024*
Participation in exercises and exams preparation (Bachelor 3 / Master 1 level)

MOOC GENIUS for high school students – Polytechnique, Palaiseau *2020 – 2021*
Co-writing of a module of the maths section

TALKS

- PGMO Days' 26 – slides (parallel session) November 2026, Palaiseau, France.
- MOP Retreat #4 – slides (group seminar) October 2026, Saarbruecken, Germany.
- PGMO Days' 25 – slides (parallel session) November 2025, Palaiseau, France.
- CEA-List DILS seminar – slides (invited by a research team) October 2025, Palaiseau, France.
- CAS seminar, slides (invited by a research team) October 2025, Paris, France.
- PhD defense at Inria Paris -- slides July 2025, Paris, France.
- ISMP 2024, slides (parallel session) July 2024, Montréal, Canada.
- NOPTA 2024 poster (summer school on nonsmoothness) April 2024, Anvers, Belgium.
- Julia Days 2023, slides October 2023, Paris, France.
- Québec colloquium of the ISM – slides (parallel session) June 2023, Sherbrooke, Canada.
- JOPT2023 – slides (parallel session) May 2023, Montréal, Canada.
- Annual days of reseach group MOA – slides October 2022, Nice, France.
- JOPT2022 – slides (parallel session) May 2022, Montréal, Canada.

GRANTS

ISM Excellence Grant (2000€)– Institute of Mathematical Sciences of Canada *2024*

Globalink Research Grant (4000€) – MITACS-Inria *2023*

Sophie Germain Academic Excellence Grant (master) (10000€) – FMJH *2020 – 2021*

TRIVIA

Software: Julia, Matlab *French*: mother tongue *English*: fluent (for work)
Hobbies: cooking, board/card/video games